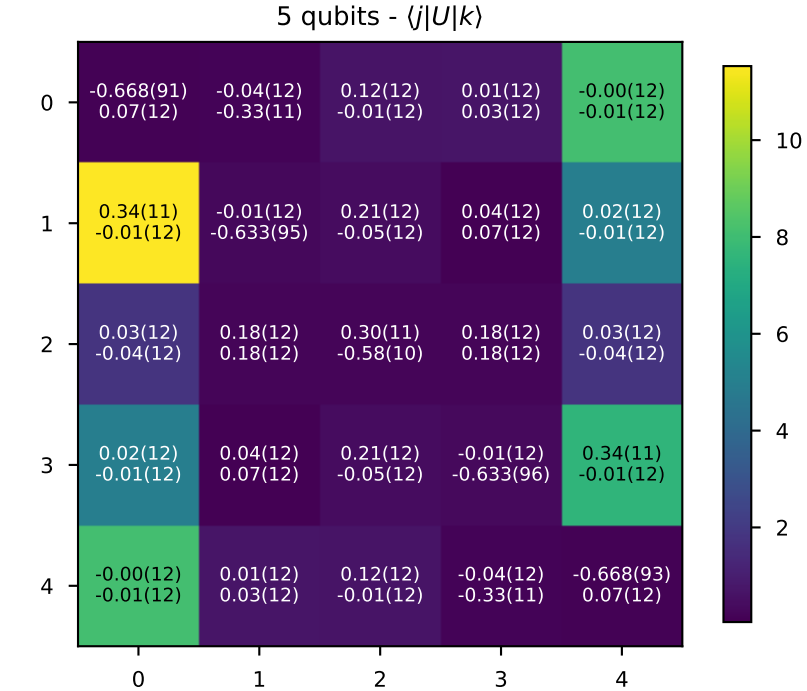
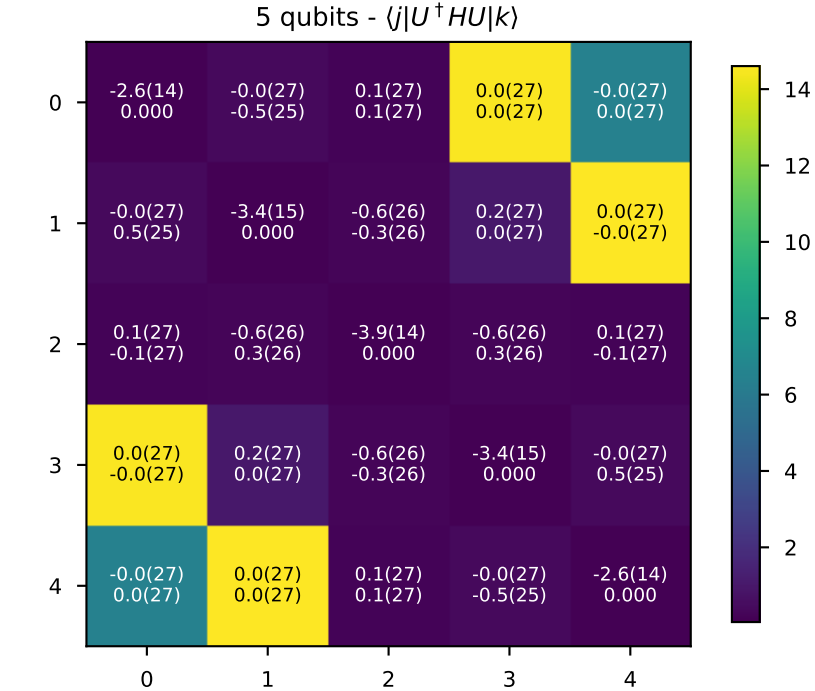
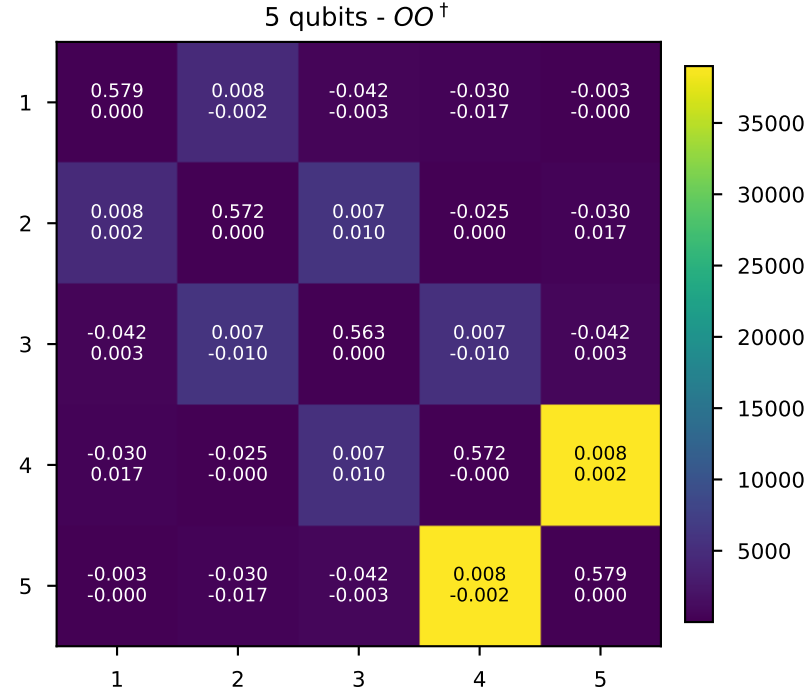
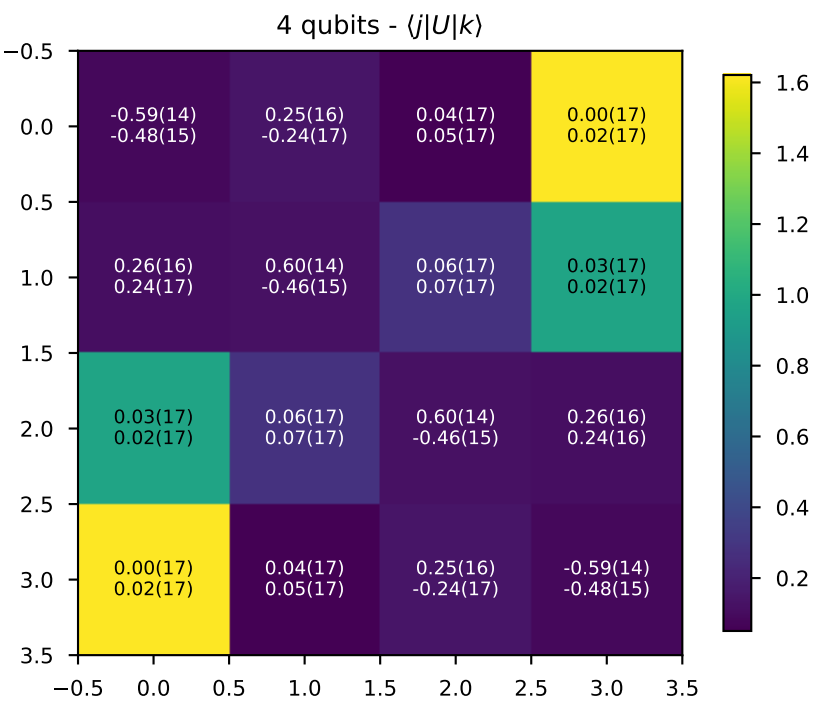
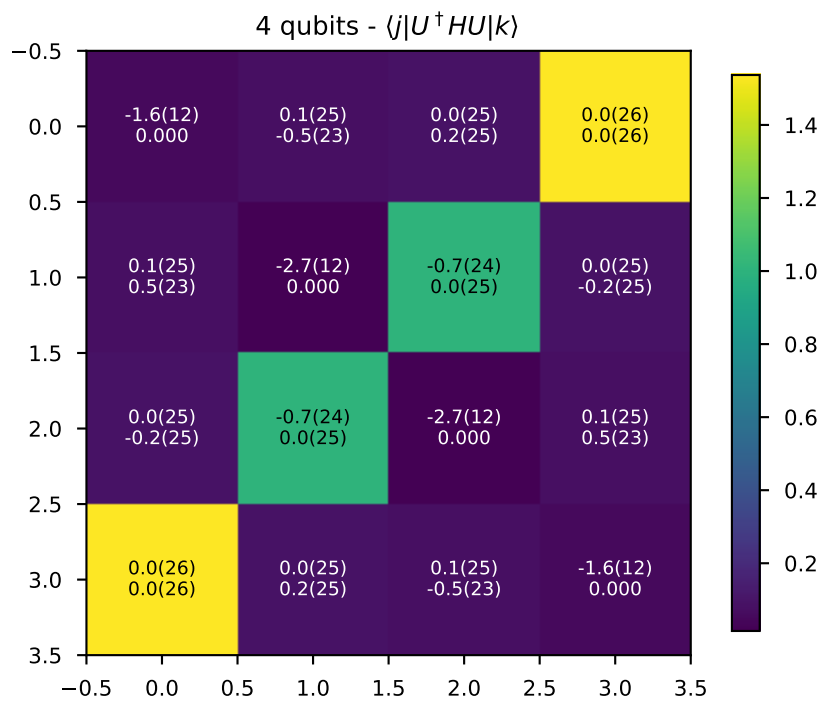
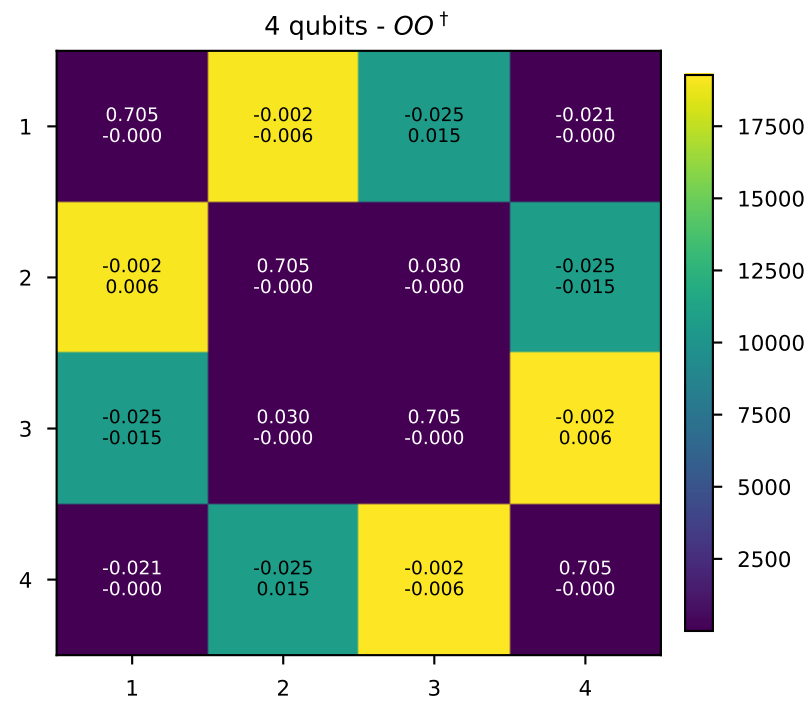

5 qubits - ⟨j|U|k⟩
-0.668(91) 0.07(12)
-0.04(12) -0.33(11)
0.12(12) -0.01(12)
0.01(12) 0.03(12)
-0.00(12) -0.01(12)
0.34(11) -0.01(12)
-0.01(12) -0.633(95)
0.21(12) -0.05(12)
0.04(12) 0.07(12)
0.02(12) -0.01(12)
0.03(12) -0.04(12)
0.18(12) 0.18(12)
0.30(11) -0.58(10)
0.18(12) 0.18(12)
0.03(12) -0.04(12)
0.02(12) -0.01(12)
0.04(12) 0.07(12)
0.21(12) -0.05(12)
-0.01(12) -0.633(96)
0.34(11) -0.01(12)
-0.00(12) -0.01(12)
0.01(12) 0.03(12)
0.12(12) -0.01(12)
-0.04(12) -0.33(11)
-0.668(93) 0.07(12)
0 1 2 3 4
2 4 6 8 10


5 qubits - ⟨j|U†HU|k⟩
-2.6(14) 0.000
-0.0(27) -0.5(25)
0.1(27) 0.1(27)
0.0(27) 0.0(27)
-0.0(27) 0.0(27)
-0.0(27) 0.5(25)
-3.4(15) 0.000
-0.6(26) -0.3(26)
0.2(27) 0.0(27)
0.0(27) -0.0(27)
0.1(27) -0.1(27)
-0.6(26) 0.3(26)
-3.9(14) 0.000
-0.6(26) 0.3(26)
0.1(27) -0.1(27)
0.0(27) -0.0(27)
0.2(27) 0.0(27)
-0.6(26) -0.3(26)
-3.4(15) 0.000
-0.0(27) 0.5(25)
-0.0(27) 0.0(27)
0.0(27) 0.0(27)
0.1(27) 0.1(27)
-0.0(27) -0.5(25)
-2.6(14) 0.000
0 1 2 3 4
2 4 6 8 10 12 14


5 qubits - OO†
0.579 0.000
0.008 -0.002
-0.042 -0.003
-0.030 -0.017
-0.003 -0.000
0.008 0.002
0.572 0.000
0.007 0.010
-0.025 0.000
-0.030 0.017
-0.042 0.003
0.007 -0.010
0.563 0.000
0.007 -0.010
-0.042 0.003
-0.030 0.017
-0.025 -0.000
0.007 0.010
0.572 -0.000
0.008 0.002
-0.003 -0.000
-0.030 -0.017
-0.042 -0.003
0.008 -0.002
0.579 0.000
1 2 3 4 5
5000 10000 15000 20000 25000 30000 35000


4 qubits - ⟨j|U|k⟩
-0.59(14) -0.48(15)
0.25(16) -0.24(17)
0.04(17) 0.05(17)
0.00(17) 0.02(17)
0.26(16) 0.24(17)
0.60(14) -0.46(15)
0.06(17) 0.07(17)
0.03(17) 0.02(17)
0.03(17) 0.02(17)
0.06(17) 0.07(17)
0.60(14) -0.46(15)
0.26(16) 0.24(16)
0.00(17) 0.02(17)
0.04(17) 0.05(17)
0.25(16) -0.24(17)
-0.59(14) -0.48(15)
−0.5 0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5
0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6


4 qubits - ⟨j|U†HU|k⟩
-1.6(12) 0.000
0.1(25) -0.5(23)
0.0(25) 0.2(25)
0.0(26) 0.0(26)
0.1(25) 0.5(23)
-2.7(12) 0.000
-0.7(24) 0.0(25)
0.0(25) -0.2(25)
0.0(25) -0.2(25)
-0.7(24) 0.0(25)
-2.7(12) 0.000
0.1(25) 0.5(23)
0.0(26) 0.0(26)
0.0(25) 0.2(25)
0.1(25) -0.5(23)
-1.6(12) 0.000
−0.5 0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5
0.2 0.4 0.6 0.8 1.0 1.2 1.4


4 qubits - OO†
0.705 -0.000
-0.002 -0.006
-0.025 0.015
-0.021 -0.000
-0.002 0.006
0.705 -0.000
0.030 -0.000
-0.025 -0.015
-0.025 -0.015
0.030 -0.000
0.705 -0.000
-0.002 0.006
-0.021 -0.000
-0.025 0.015
-0.002 -0.006
0.705 -0.000
1 2 3 4
2500 5000 7500 10000 12500 15000 17500